Action Differences, Fluctuations and Operational Bounds

Control results for a research programme on mechanical gaps

navstokgap research working notes

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Abstract

For a particle launched perpendicular to a constant force, we derive the exact chord–curve action difference $\Delta S = F^2T^3/(24m) = T\delta E/12$ and relate it to the geometric error area. Positive fixed-endpoint action differences accumulate at zero; a Taylor argument extends this result to nondegenerate variations. For constant force, the Dirichlet Jacobi operator has a positive lowest eigenvalue, separating normalized fluctuation cost from action amplitude. The free quantum kernel converges to a delta with a second-derivative short-time correction. Finally, within a specified quantum two-arm measurement model, we derive an exact action-resolution threshold for N independent copies and target success probability p. For fixed $1/2 < p < 1$, the threshold decreases as $N^{-1/2}$. These results connect trajectory geometry, fluctuation spectra and operational resolution in a tractable setting for studying gap formation.

1 Action values, fluctuations and resolution

The motivating question is whether limitations on Newtonian trajectories can explain a nonzero action scale, and whether a useful toy model can isolate the formation of a gap. We write $h = 2\pi\hbar$ for Planck's constants, η for a path variation, and J for a second-variation operator. An additional field χ would enter through a specified extended action $S[q, \chi]$.

For a path class $\mathcal{P}$ and reference q_*, one possible action-gap question is

$$g_S = \inf\{|S[q] - S[q_*]| : q \in \mathcal{P},\ S[q] \neq S[q_*]\} > 0.$$

Assume the set is nonempty and specify the path class, topology, endpoints and duration. The corresponding Hamiltonian gap question is $\mathrm{spec}(H) \cap (E_0, E_0 + \Delta) = \varnothing$ for some $\Delta > 0$. Here E_0 is the ground energy of a specified self-adjoint Hamiltonian. The path-space infimum and spectral interval give two precise meanings of a gap.

Newton's geometry motivates the area comparison [7, 8]. NATP00385 [6] supplies Newton's later account of analytic discovery and synthetic presentation. The following calculation expresses the geometric comparison through Hamilton's action.

2 Constant force: geometry and action

Fix $m, F, T, v_0 > 0$, set $V(y) = -Fy$, and impose $x(0) = y(0) = \dot{y}(0) = 0$, $\dot{x}(0) = v_0$. Newton's equations give

$$x_*(t) = v_0 t, \qquad y_*(t) = \frac{Ft^2}{2m}, \qquad 0 \le t \le T.$$

The kinetic-energy gain is $\delta E = F^2T^2/(2m) = -\Delta V$; total mechanical energy is conserved. The triangle between initial tangent, endpoint vertical and chord has area $v_0FT^3/(4m)$. The chord–curve lens has one third of this:

$$\mathcal{A}_{\text{lens}} = \frac{v_0 F T^3}{12m}.$$

These are frame-dependent areas in configuration space. The factor relating the lens area to action will follow from a matched-endpoint variation.

Hold $x = x_*$ fixed and consider the same-endpoint paths $y = y_* + \eta$ with $\eta \in H_0^1(0,T)$, the Sobolev space with square-integrable weak derivative and zero endpoint trace. Define

$$S[x,y] = \int_0^T \left[\frac{m}{2}(\dot{x}^2 + \dot{y}^2) + Fy\right] \mathrm{d}t.$$

The linear variation vanishes by integration by parts and $m\ddot{y}_* = F$:

$$S[x_*, y_* + \eta] - S[x_*, y_*] = \frac{m}{2}\int_0^T \dot{\eta}^2 \,\mathrm{d}t. \tag{1}$$

The straight chord $y_{\text{ch}} = FTt/(2m)$ is an admissible off-shell comparison path. Substitution into (1) gives

$$\Delta S_{\text{ch},*} = \frac{F^2 T^3}{24m} = \frac{F}{2v_0}\mathcal{A}_{\text{lens}} = \frac{T\delta E}{12}. \tag{2}$$

Adding the same total time derivative $\mathrm{d}G(q,t)/\mathrm{d}t$ to both Lagrangians leaves this difference unchanged because configuration endpoints and duration coincide. A velocity-dependent endpoint function requires additional endpoint data.

Proposition 1 (Zero infimum of action differences)**.** *For this fixed-endpoint path class the infimum of positive action differences from the classical path is zero, although the classical path is the unique minimizer within the class with fixed horizontal motion.*

Proof. If $\eta \neq 0$ in H_0^1, its derivative cannot vanish almost everywhere, so (1) is positive. For $\eta_a(t) = at(T-t)$ it equals

$$\Delta S(a) = \frac{ma^2T^3}{6}.$$

Any prescribed positive bound is violated by taking a sufficiently small nonzero a. The zero variation alone gives equality in (1). □

Proposition 2 (Local obstruction along a variation)**.** *Let q_a be admissible for all a in an open interval about zero, and suppose $f(a) = S[q_a]$ is twice continuously differentiable there, with $f'(0) = 0$ and $f''(0) = b \neq 0$. Then arbitrarily small nonzero absolute action differences $|S[q_a] - S[q_0]|$ occur. If $b > 0$, these differences can be taken positive without absolute values.*

Proof. Taylor's formula gives $f(a) - f(0) = ba^2/2 + o(a^2)$. For all sufficiently small nonzero a, its magnitude lies between $|b|a^2/4$ and $3|b|a^2/4$ and its sign is the sign of b. It is nonzero and tends to zero. □

Locality here is initially in the parameter a. To conclude that these actions occur in every path-space neighbourhood of q_0, additionally require $q_a \to q_0$ in the specified path topology. The polynomial family above converges in H^1 (and in C^1).

The admissible small-amplitude family is the mechanism behind both propositions. Discrete path classes, topological sectors and operational equivalence relations offer concrete ways to change this admissibility question. Kepler collisions pose a separate IVP question about continuation through a singular force.

3 Jacobi fields and normalized fluctuation gaps

For $S[q] = \int_0^T (m\dot{q}^2/2 - V(q))\mathrm{d}t$ about a smooth classical solution, the second variation is

$$Q[\eta] = \int_0^T \left[m\dot{\eta}^2 - V''(q_*)\eta^2\right]\mathrm{d}t, \qquad J = -m\frac{\mathrm{d}^2}{\mathrm{d}t^2} - V''(q_*).$$

For smooth bounded $V''(q_*(t))$ take the Dirichlet operator domain $H^2(0,T) \cap H_0^1(0,T)$ in $L^2(0,T)$. A variation through classical solutions satisfies the Jacobi equation $J\eta = 0$. The full variational path class also contains off-shell fluctuations. For constant force a Jacobi field is $A + Bt$, so fixed zero endpoints force it to vanish. Equation (1) nevertheless admits nonzero fluctuations.

The second-variation framework is classical [1, §8.2]. For constant force the Dirichlet eigenfunctions are $\sin(n\pi t/T)$ with eigenvalues $\lambda_n = m(n\pi/T)^2$, $n \geq 1$. Thus

$$Q[\eta] \geq \frac{m\pi^2}{T^2}\|\eta\|_{L^2}^2.$$

This is coercivity relative to the L^2 norm: rescaling η rescales Q quadratically. The eigenvalue λ_n has units mass/time^2, while $\|\eta\|_{L^2}^2$ has units length^2 time, producing action.

For the oscillator $V = m\omega^2 q^2/2$, the corresponding eigenvalues become $m[(n\pi/T)^2 - \omega^2]$. Zero modes occur at $T = n\pi/\omega$; positivity of all modes fails after the first such time. This explicit calculation is a useful comparison with the quantum oscillator Hamiltonian. The companion working draft *A Gap Laboratory: Oscillators, Circles and Winding Sectors* derives both spectra, their domains and their distinct gap-closing limits.

4 What distribution is the oscillatory kernel?

The quantum path-phase input is $\exp(iS/\hbar)$, with time-integrated action, as in the Feynman formulation [3, 4]. To study its distributional meaning, specify a kernel, normalization and limiting parameter. The free one-dimensional Schrödinger kernel provides an exact example [9, §7.3, (7.30)]. The expansion below is its Fourier consequence. For $\hbar, m, \tau > 0$ let

$$K_\tau(z) = \left(\frac{m}{2\pi i\hbar\tau}\right)^{1/2} \exp\left(\frac{imz^2}{2\hbar\tau}\right),$$

with the usual branch $1/\sqrt{i} = e^{-i\pi/4}$ and oscillatory/tempered-distribution interpretation. Use Fourier transform $\widehat{f}(k) = \int e^{-ikz} f(z)\mathrm{d}z$. Equivalently this kernel is defined by

$$\widehat{K_\tau}(k) = \exp\left(-\frac{i\hbar\tau k^2}{2m}\right).$$

For each Schwartz test function, dominated convergence in Fourier space implies

$$K_\tau \longrightarrow \delta \quad (\tau \downarrow 0,\ \hbar > 0 \text{ fixed}). \tag{3}$$

Taylor expansion of the multiplier, using $|e^{-iu} - 1 + iu| \leq u^2/2$ for real u, further gives

$$K_\tau = \delta + \frac{i\hbar\tau}{2m}\delta'' + O(\tau^2) \quad \text{in the sense of pairing with each fixed Schwartz test function.}$$

The remainder estimate follows by integrating k^4 times the absolute value of the transformed test function. The even free kernel gives δ at leading order and δ'' at first order in time. Differentiating the kernel with respect to z instead gives a family converging to δ'. A Fourier constraint integral provides another construction: $\int e^{i\lambda z/\hbar}\mathrm{d}\lambda/(2\pi\hbar) = \delta(z)$. Substitution of a nonlinear constraint requires further regularity and Jacobian conditions.

The limit above is short time at fixed $\hbar$. A semiclassical analysis instead varies $\hbar$; each limit has its own test functions and convergence statement.

5 A conditional operational action threshold

Assume the quantum state/measurement framework, a specified positive $\hbar$, and a coherently controlled two-arm system realizing a relative phase $\phi = \Delta S/\hbar$. The relative arm phase encodes two known hypotheses with equal priors:

$$|\psi_0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |\psi_\phi\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}.$$

Allow $N \geq 1$ independent identical copies and any joint binary quantum measurement. These assumptions define the resource budget and the hypothesis test.

The following is a pure-state specialization of the Holevo–Helstrom theorem [10, Theorem 3.4], followed by the specified action-phase encoding.

Proposition 3 (Finite-copy bound). *The optimal equal-prior success probability is*

$$P_N(\phi) = \frac{1}{2}\left(1 + \sqrt{1 - \cos^{2N}(\phi/2)}\right). \tag{4}$$

For $1/2 < p \leq 1$, in the first phase lobe $|\phi| \leq \pi$, success at least p is possible if and only if

$$|\Delta S| \geq d_{N,p} := 2\hbar \arccos\left([4p(1-p)]^{1/(2N)}\right). \tag{5}$$

For each fixed $1/2 < p < 1$, $d_{N,p} \to 0$ as $N \to \infty$. For $p = 1$, the finite-copy threshold instead remains $\pi\hbar$.

Proof. Two normalized pure states with overlap magnitude r have density-matrix difference D with eigenvalues $\pm\sqrt{1-r^2}$ on their span (and zero outside). Indeed $\operatorname{tr} D = 0$ and $\operatorname{tr} D^2 = 2(1-r^2)$. A binary measurement effect $0 \leq E \leq I$ gives success $1/2 + \operatorname{tr}(ED)/2$, maximized by projecting onto the positive eigenspace. The maximum is $(1+\sqrt{1-r^2})/2$. Here a single-copy overlap has magnitude $|\cos(\phi/2)|$, and tensor products raise it to the Nth power, proving (4). Within the first lobe cosine is nonnegative and monotone in $|\phi|$, so solving $P_N \geq p$ gives (5). For $0 < 4p(1-p) < 1$ its $1/(2N)$ power tends to one. At $p = 1$ it is zero for every finite N, giving the stated exception. □

For example $N = 1$, $p = 3/4$ gives $d_{1,3/4} = \pi\hbar/3$. For fixed $1/2 < p < 1$ the asymptotic threshold is

$$d_{N,p} \sim 2\hbar\sqrt{\frac{-\log(4p(1-p))}{N}}.$$

The threshold is positive at fixed resources and accuracy. Its dependence on N quantifies the improvement in resolution from repeated preparations. Phase periodicity explains the first-lobe restriction: at $\phi = 2\pi n$ the hypotheses coincide again.

If an apparatus realizes the particular relative action (2), with all control and recombination phases included, its first-lobe threshold would be

$$T \geq \left(\frac{24 m d_{N,p}}{F^2}\right)^{1/3}.$$

The upper first-lobe condition $F^2T^3/(24m\hbar) \leq \pi$ must be retained. Realizing the off-shell chord as an experimental arm requires a control force; the displayed substitution applies when the full apparatus has the specified relative action.

6 Finite propagation and the field-theory companion

For the free relativistic Lagrangian $L = -mc^2\sqrt{1-\dot{q}^2/c^2}$, consider the fixed-endpoint family $q_a = at(T-t)$, with both endpoints zero. If $|a|T < c$, the path has a strict speed margin. Relative to the rest path its action excess is positive for $a \neq 0$ and

$$\Delta S(a) = mc^2 \int_0^T \left(1 - \sqrt{1 - a^2(T-2t)^2/c^2}\right) \mathrm{d}t = \frac{ma^2T^3}{6} + O(a^4) \longrightarrow 0.$$

Thus positive action differences accumulate at zero within the strict speed margin, just as in the Newtonian example. This gives a kinematic comparison at finite c.

Navier–Stokes and Yang–Mills remain complementary comparisons, with their official targets kept intact [2, 5]. A diffusion or Dirichlet coercivity bound can depend on time, domain and normalization; a physical quantum mass gap concerns a different operator and limiting problem. Any future bridge must identify the spaces, physical versus auxiliary time, estimates and limits it preserves. The fluid target is regularity of evolution; the field-theory target includes construction of the physical quantum theory and its vacuum gap.

7 Scope, reproduction and next experiments

The results concern the stated mechanical path classes and quantum measurement model. Hilbert states, the Born rule and positive $\hbar$ are physical inputs to Section 5. Deriving these premises belongs to the reconstruction branch of the companion programme. The proofs received internal review; novelty assessment and external review are publication-stage tasks.

The repository's `make check` verifies the exact polynomial action identities, the oscillator-mode residual, the Fourier expansion coefficient and elementary discrimination cross-checks. `make papers` rebuilds this note and the companion programme PDF. The formalisation queue begins with the small-amplitude polynomial lemma.

The companion *Time Refinement and an Action Scale* develops the free Gaussian blocking test. Next steps audit the regulator proposal and quantum reconstruction axioms, alongside central-potential models and the historical corpus inventory, with sequential source audits. A successful extension must identify which premise excludes the classical small-amplitude families, or else choose an operational or spectral gap instead and state the change of target. Deriving a theory's form, forcing a nonzero universal scale and determining its measured numerical value are separate tasks.

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